

Hyunsoo Kim

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EDUCATION

The University of Texas at Austin

Austin, TX

Ph.D. in Electrical and Computer Engineering

Sep 2026 – Present

- Co-advised by Prof. Guandao Yang and Prof. Diana Marculescu
- Research: Video-based world models, autoregressive video generation, and efficient inference for world models

Korea University

Seoul, Republic of Korea

M.S. in Artificial Intelligence

Sep 2024 – Aug 2026

- Co-advised by Prof. Donghyun Kim and Prof. Suhyun Kim
- Research: Diffusion models and generative model alignment

Inha University

Incheon, Republic of Korea

B.S. in Statistics and Software Engineering

Mar 2018 – Aug 2024

- Two-year break for military service (Jul 2019 – Feb 2021)

RESEARCH INTERESTS

Video-based world models (long-horizon, action- and subject-conditioned video generation)

Autoregressive video generation (streaming generation, KV-cache conditioning, long-term consistency)

Efficient inference for world models (speculative decoding, few-step sampling)

Generative models (diffusion models, flow matching, post-training)

PREPRINTS

[P1] Custom Forcing: Training-Free Subject Customization for AR Video Generation

Preprint

- Yunseung Ok*, **Hyunsoo Kim***, Minseo Kim, Suhyun Kim[†] (*: equal contribution, †: corresponding author)

PUBLICATIONS

[C5] Consistency-Preserving Concept Erasure via Unsafe-Safe Pairing (PAIR)

NeurIPS 2026

- Yongwoo Kim*, Sungmin Cha*, **Hyunsoo Kim**, Jaewon Lee, Donghyun Kim (*: equal first authorship)

[C4] Correlation-Weighted Multi-Reward Optimization for Compositional Generation

ECCV 2026

- Jungmyung Wi, **Hyunsoo Kim**, Donghyun Kim

[C3] Self-Improving Diffusion Classifiers with Minority Preference Optimization

ECCV 2026

- **Hyunsoo Kim***, Jungmyung Wi*, Soobin Um, Donghyun Kim[†], Suhyun Kim[†] (*: equal contribution, †: equal advising)

[C2] Diffusion-assisted Data-free Image Synthesis with Domain-Class Alignment

ICML 2025

- Yujin Kim*, **Hyunsoo Kim***, Hyunwoo J. Kim, Suhyun Kim (*: equal contribution)

[C1] Difference Inversion: Interpolate and Isolate the Difference with Token Consistency

CVPR 2025

- **Hyunsoo Kim**, Donghyun Kim[†], Suhyun Kim[†] (†: equal advising)

TEACHING

Teaching Assistant, ECE 381V: Advanced Topics in Computer Vision, The University of Texas at Austin Fall 2026

EXPERIENCE

Korea Institute of Science and Technology (KIST) <i>Graduate Researcher (Advisor: Suhyun Kim)</i>	Seoul, Republic of Korea <i>Sep 2024 – Aug 2026</i>
• Led research on diffusion-based generation, generative model alignment, and diffusion classifiers • Developed training pipelines for diffusion model fine-tuning with RLHF/GRPO on multi-GPU clusters	
Korea Institute of Science and Technology (KIST) <i>Undergraduate Research Intern (Advisor: Suhyun Kim)</i>	Seoul, Republic of Korea <i>Mar 2024 – Jun 2024</i>
• Researched data-free image synthesis with text-to-image diffusion models	
Medical Image Innovation Lab, Seoul National University <i>Undergraduate Intern (Advisor: Wonjin Lee)</i>	Seoul, Republic of Korea <i>Dec 2023 – Feb 2024</i>
• Researched diffusion-based conditional generation and inpainting for medical imaging	

SKILLS

Python, PyTorch, Hugging Face Diffusers, DeepSpeed, SLURM / HPC clusters, Git, Docker